

For a set $X \subseteq V(G)$, the *(open) neighborhood* of X , denoted by $N_G(X)$, is the set $\{v \in V(G) \setminus X : v \text{ is adjacent to a vertex in } X\}$. The *closed neighborhood* of X is the set $N_G[X] = N_G(X) \cup X$. We simply use $N(v)$, $N[v]$, $N(X)$, and $N[X]$ to denote $N_G(v)$, $N_G[v]$, $N_G(X)$, and $N_G[X]$, respectively when the context of the graph G is clear. We use $G[X]$ to denote the subgraph of G induced by the vertices in X . A vertex v dominates itself and all its neighbors. A set $X \subseteq V(G)$ is called a *clique* of G if every pair of vertices of X are adjacent. For disjoint subsets X and Y of $V(G)$, $G[X, Y]$ denotes the set of edges $\{e \in E(G) : e \text{ has one endpoint in } X \text{ and the other endpoint in } Y\}$. If the graph G is clear from the context, we write $[X, Y]$ rather than $G[X, Y]$. We say that $G[X, Y]$ is complete if every vertex in X is adjacent to every vertex in Y in G . For a given positive integer k , we use the notation $[k]$ to denote the set $\{1, \dots, k\}$.

For a set $D \subseteq V(G)$ and a vertex $u \in D$, the *D -external private neighborhood* of u , denoted by $\text{epn}(u, D)$, is the set $\{v \in V(G) \setminus D : N(v) \cap D = \{u\}\}$. A vertex in $\text{epn}(u, D)$ is a *D -external private neighbor* of u . We say that a vertex $v \in V(G) \setminus D$ is *D -defended* if there exists a vertex $u \in N(v) \cap D$ such that $(D \setminus \{u\}) \cup \{v\}$ is a dominating set of G . Specifically, we say that $v \in V(G) \setminus D$ is *D -defended by a vertex u* if $u \in N(v) \cap D$ and $(D \setminus \{u\}) \cup \{v\}$ is a dominating set of G . We note that D is a secure dominating set of G if and only if every vertex in $V(G) \setminus D$ is D -defended. We also note that if D is a dominating set of G and $\text{epn}(u, D) = \emptyset$ for some $u \in D$, then every neighbor of u in $V(G) \setminus D$ is D -defended by u . We state this observation formally as follows.

Observation 2. *If D is a dominating set of a graph G and $\text{epn}(u, D) = \emptyset$ for some vertex $u \in D$, then every neighbor of u that belongs to $V(G) \setminus D$ is D -defended by u .*

As a consequence of Observation 2, if D is a dominating set of G and $\text{epn}(u, D) = \emptyset$ for every $u \in D$, then every vertex in $V(G) \setminus D$ is D -defended, implying that D is a secure dominating set of G . For a given dominating set D of G , we partition D as $A_D \cup B_D$, where the sets A_D and B_D are defined as follows.

- $A_D = \{u \in D : \text{there exists a neighbor of } u \text{ in } V(G) \setminus D \text{ that is not } D\text{-defended.}\}$ ($\mathcal{P}_1$)
- $B_D = D \setminus A_D$.

We now prove the following useful lemma. We remark that some of the facts that appear in the following lemma have appeared implicitly in several papers (see [4, 15]).

Lemma 1. *Let D be a dominating set of G and $A_D \cup B_D$ be a partition of D as defined in ($\mathcal{P}_1$).*

- (a) *If $A_D = \emptyset$, then D is a secure dominating set of G .*
- (b) *If a vertex v in $V(G) \setminus D$ is not D -defended, then $N(v) \cap D \subseteq A_D$.*
- (c) *If u is a vertex in A_D , then $\text{epn}(u, D) \neq \emptyset$.*
- (d) *If a vertex v in $V(G) \setminus D$ is not D -defended, then for every neighbor u of v in D , the vertex v has a non-neighbor in $\text{epn}(u, D)$, that is, $\text{epn}(u, D) \setminus N(v) \neq \emptyset$.*
- (e) *If D' is a superset of D , then $A_{D'} \subseteq A_D$ and $B_D \subseteq B_{D'}$, where $A_{D'} \cup B_{D'}$ is a partition of D' as defined in ($\mathcal{P}_1$).*

Proof. Parts (a) and (b) follow by the definition of the sets A_D and B_D , while part (c) follows by Observation 2.